AMD DEVELOPER HACKATHON · ACT III

SIMON

Verifiable LLM inference on hardware that you do not control

Nodes sign what they computed. Anyone can verify it offline.

github.com/Szpili/SIMON · MIT · Rust · libp2p

THE PROBLEM

A signature proves who wrote the number.
It never proves the number is true.

Every decentralized inference network hands you a signed claim and calls it a proof. Ours started the same way. The interesting engineering is in the gap between those two words.

What a receipt proves and what it does not

Proved today

A registered node signed this exact answer as the result of this job.

- Ed25519 signature over a commitment to the output text
- Bound to the job id. A valid receipt from another run cannot be reused

Not proved

That the declared model generated it.

- A malicious node can sign arbitrary text with a correct digest
- Token counts and timings are signed declarations, not independent measurements

Binding model + prompt + execution + output needs an execution audit. It is not implemented, and the repository says so.

LIVE DEMO

Try to forge the receipt yourself

Tamper with the answer

Change one byte after signing.

REJECTED: the signature breaks

Replay another job

A correctly signed receipt from a different job.

REJECTED: not proof of this work

Swap the text

Real signature, substituted answer.

REJECTED: the digest covers the text

No simulated mode. If no node is reachable, the demo says so instead of a canned result.

MEASURED, NOT PROJECTED

An audit is cheap enough to pay for

An audit is one forward pass. The job must also decode, and decoding is ~55× slower per token.

Prompt	Output	Job	Audit	Audit as % of job
500	50	7.9 s	0.63 s	**8.0%**
2016	200	18.8 s	1.86 s	**9.9%**
2011	500	87.1 s	2.07 s	**2.4%**
6037	500	69.1 s	5.12 s	**7.4%**

Lower bound: this is the forward pass alone, without activation extraction and hashing. Raw data is in the repository.

Heterogeneous on purpose, and it shows

Two GPUs, two serving stacks, two models, one network. The engine and the model on each result come from the signed receipt, not from a label that we wrote beside it.

RTX 3090 · vLLM 0.27.1

Qwen3.8-27B · full context admission gate

64.94 tok/s

RTX 3080 Ti · Ollama 0.34.0

Bielik 11B · admission gate disabled, and the node says so

6.50 tok/s

A node that cannot measure its own context budget declares that. It does not pretend.

Does an honest AMD node compute the same thing as an honest NVIDIA one?

If the answer is no, a naive fingerprint comparison slashes honest participants. Every cross-vendor network has this problem, whether it noticed or not.

Already done

The CUDA baseline is recorded. The notebook ships with it embedded, so it returns a verdict in one session.

Needs MI300X

The ROCm arm under vLLM. Our backend speaks an OpenAI-compatible API, so this is configuration, not a rewrite.

We will publish it even if the result is inconvenient for us.

Three of our own claims, withdrawn in public

The receipt did not bind the answer

Found during a test of whether we can count tokens at all

The digest covered metadata. The text traveled unsigned.

A constant pretending to be a measurement

It survived the code, a comment that claimed "hard measurement", and a demo that showed one number on two GPUs

Throughput was tokens / 65.0, the figure of one card, applied to every node.

A test that the whitepaper promised and that we never wrote

Corrected in both languages. The ledger now gives it somewhere to be true

Both papers said that the client tests whether a receipt is single-use. It did not.

Status, without rounding up

Capability	State
Output binding	**implemented, tested**
Offline receipt verification by a third party	**implemented, tested**
Chunking, map-reduce, agent-to-agent chaining	**working**
Token counts recomputed independently	**partial: backends expose no token ids**
Execution correctness (TopLoc-style audit)	**not implemented**
Automatic slashing	**off by decision: no defensible threshold**
Wash-compute resistance	**unsolved**

129 tests green · CI on every push, with a guard against constants that pose as measurements

Idle capacity turned into burst capacity, with a receipt that you can verify

This is not "earn while you sleep". At 350 W, a night of full load is worth less in commodity tokens than the electricity. The value is work that must not leave your machines, and results that you can verify. It is also five GPUs at once on the morning you need them.

Code, MIT

github.com/Szpili/SIMON

Also in the repository

Whitepaper (EN and PL) · security log · ROCm notebook